

Q=1) WAP to find max and min elements in an array

Ans)

```
#include<stdio.h>
int main()
{
    int arr[100],size,i,max,min;
    printf("Enter the size of array: \n");
    scanf("%d",&size);
    for ( i = 0; i <size; i++)
    {
        scanf("%d",&arr[i]);
    }
    max=arr[0];
    for ( i = 0; i <size; i++)
    {
        if (arr[i]>max)
        {
            max=arr[i];
        }
    }
    printf("The max value of array is %d\n",max);
    min=arr[0];
    for ( i = 0; i <size; i++)
    {
        if (arr[i]<min)
        {
            min=arr[i];
        }
    }
    printf("The min value of array is %d",min);
    return 0;
}
```

Enter the size of array:

5

10

20

30

40

50

The max value of array is 50The min value of array is 10

Q=2) WAP to multiply two matrices using 2D arrays

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```

int a[3][3], b[3][3], c[3][3], i, j, k;
printf("Enter first matrix i.e a \n");
for (i = 0; i < 3; i++)
{
    for (j = 0; j < 3; j++)
    {
        scanf("%d", &a[i][j]);
    }
    printf("Enter second matrix i.e b \n");
    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
        {
            scanf("%d", &b[i][j]);

        }
        for ( i = 0; i < 3; i++)
        {
            for (j = 0; j< 3; j++)
            {
                c[i][j]=0;
                for ( k = 0; k< 3; k++)
                {
                    c[i][j]=c[i][j] +a[i][k]*b[i][k];
                }
            }
        }
        printf("The result is :\n\n");

    }

    for ( i = 0; i < 3; i++)
    {
        for ( j= 0; j < 3; j++)
        {
            printf("%d\t",c[3][3]);
        }

    } return 0;
}

```

```

}
Enter first matrix i.e a
1 2 3 4 5 6 7 8 9
Enter second matrix i.e b
9 8 7 6 5 4 3 2 1
The result is :

```

```

6 6 6 6 6 6 _ 6 6 6

```

Q 3) WAP to check the square matrix entered by the user is symmetric matrix or not, using 2D array

```
#include <stdio.h>
int main()
{
    int a[3][3], b[3][3], i, f = 0, j;
    printf("enter the matrix a as:\n");
    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }
    printf("Enter the second matrix as \n");
    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
        {
            b[i][j] = a[j][i];
        }
    }

    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 3; j++)
        {
            if (a[i][j] != b[i][j])
            {
                f = 1;
                break;
            }
        }
        if (f == 0)
        {
            printf("the matrices are symmetric");
        }
        else
        {
            printf("The matrices are not symmetric");
        }
        return 0;
    }
}
```

enter the matrix a as:
1 2 3 4 5 6 7 8 9
Enter the second matrix as
The matrices are not symmetric

Q 4) WAP to find occurrence of a number in an integer array

```
#include <stdio.h>
int main()
{
    int arr[10], key, i, count=0;
    printf(" Enter the matrix as:\n");
    for (i = 0; i < 10; i++)
    {
        scanf("%d", &arr[i]);
    }
    printf("Enter the no. you want to search:\n");
    scanf("%d", key);
    for (i = 0; i < 10; i++)
    {
        if (arr[i] == key)
        {
            printf("%d is found ain array %d", key,i);
            count++;
        }
    }
    printf("%d has appeared",key);
    return 0;
}
```

Enter the matrix as:
;4 3 5 5 2 1 6 7 8 9
Enter the no. you want to search:
5